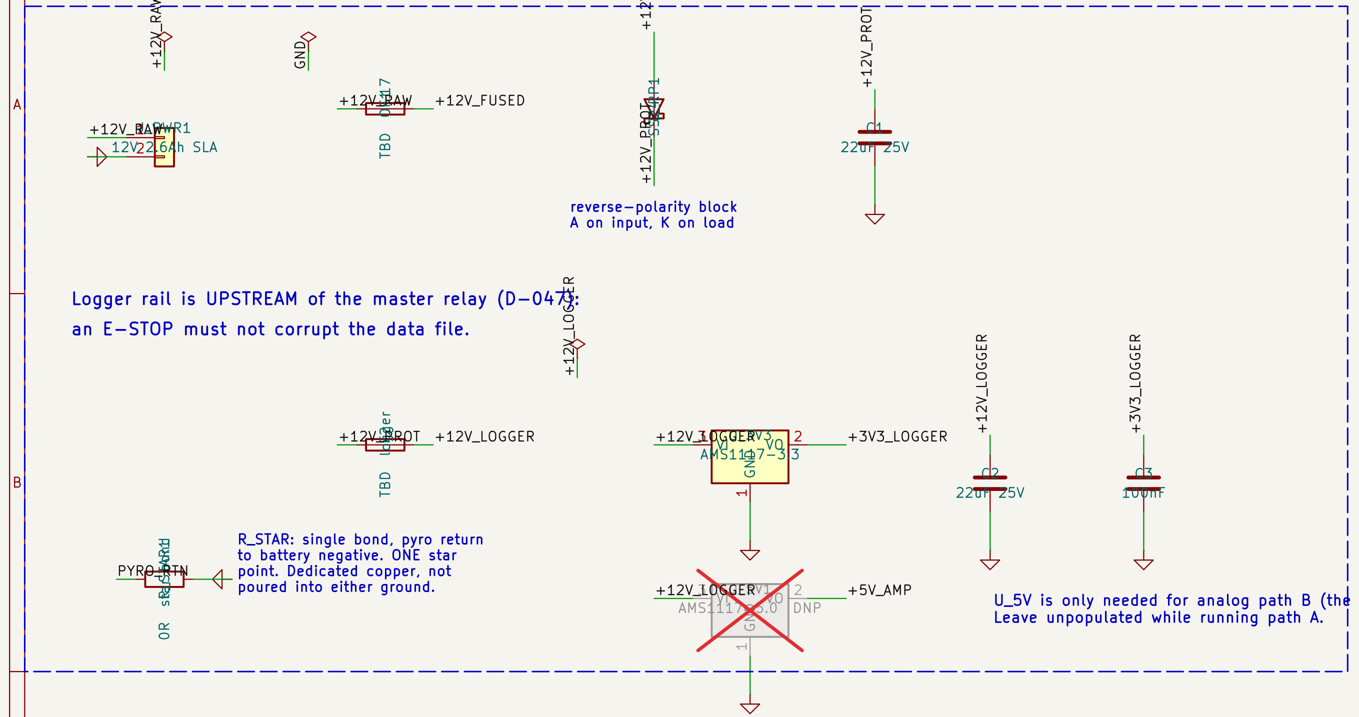
1. POWER & PROTECTION


+12V_RAW
GND
+12V_RAW
PWR1
12V 2.6Ah SLA
+12V_RAW
+12V_FUSED
TBD
+12V_FUSED
+12V_PROT
+12V_PROT
C1
22uF 25V
+12V_LOGGER
+12V_PROT
+12V_LOGGER
TBD
+12V_LOGGER
+3V3_LOGGER
AMS1117-3.3
+12V_LOGGER
C2
22uF 25V
+3V3_LOGGER
C3
100nF
PYRO_RTN
OR
+12V_LOGGER
+5V_AMP
DNP

reverse-polarity block
A on input, K on load

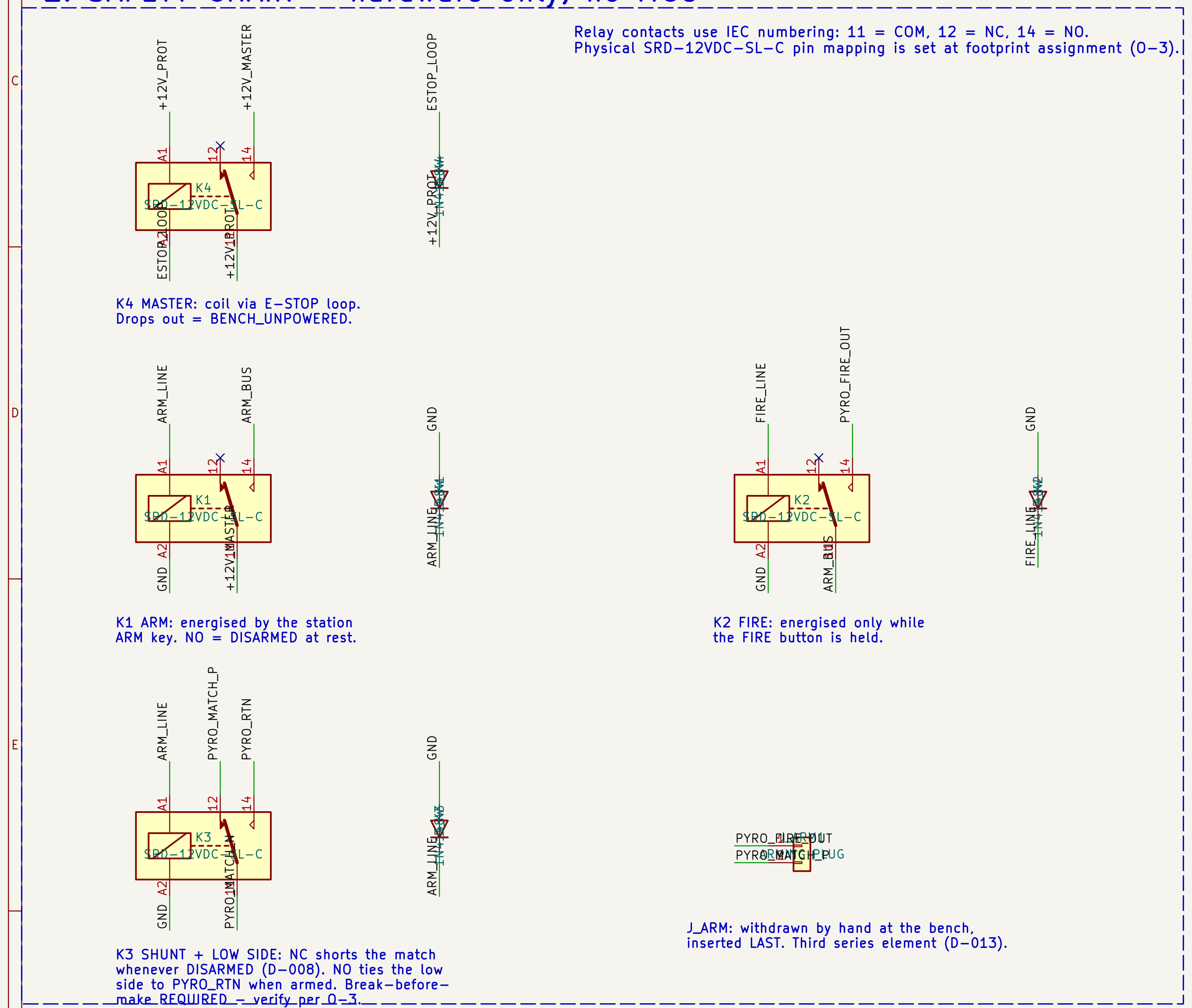
Logger rail is UPSTREAM of the master relay (D-047):
an E-STOP must not corrupt the data file.

R_STAR: single bond, pyro return
to battery negative. ONE star
point. Dedicated copper, not
poured into either ground.

U_5V is only needed for analog path B (the AD620).
Leave unpopulated while running path A.

2. SAFETY CHAIN – hardware only, no MCU

Relay contacts use IEC numbering: 11 = COM, 12 = NC, 14 = NO.
Physical SRD-12VDC-SL-C pin mapping is set at footprint assignment (O-3).


+12V_PROT
+12V_MASTER
K4
SRD-12VDC-SL-C
ESTOP_LOOP
+12V_PROT
ESTOP_LOOP
+12V_PROT
ARM_LINE
ARM_BUS
K1
SRD-12VDC-SL-C
GND
+12V_MASTER
GND
ARM_LINE
FIRE_LINE
PYRO_FIRE_OUT
K2
SRD-12VDC-SL-C
GND
ARM_BUS
GND
FIRE_LINE
ARM_LINE
PYRO_MATCH_P
PYRO_RTN
K3
SRD-12VDC-SL-C
GND
PYRO_MATCH_N
GND
ARM_LINE
PYRO_FIRE_OUT
PYRO_MATCH_P

K4 MASTER: coil via E-STOP loop.
Drops out = BENCH_UNPOWERED.

K1 ARM: energised by the station
ARM key. NO = DISARMED at rest.

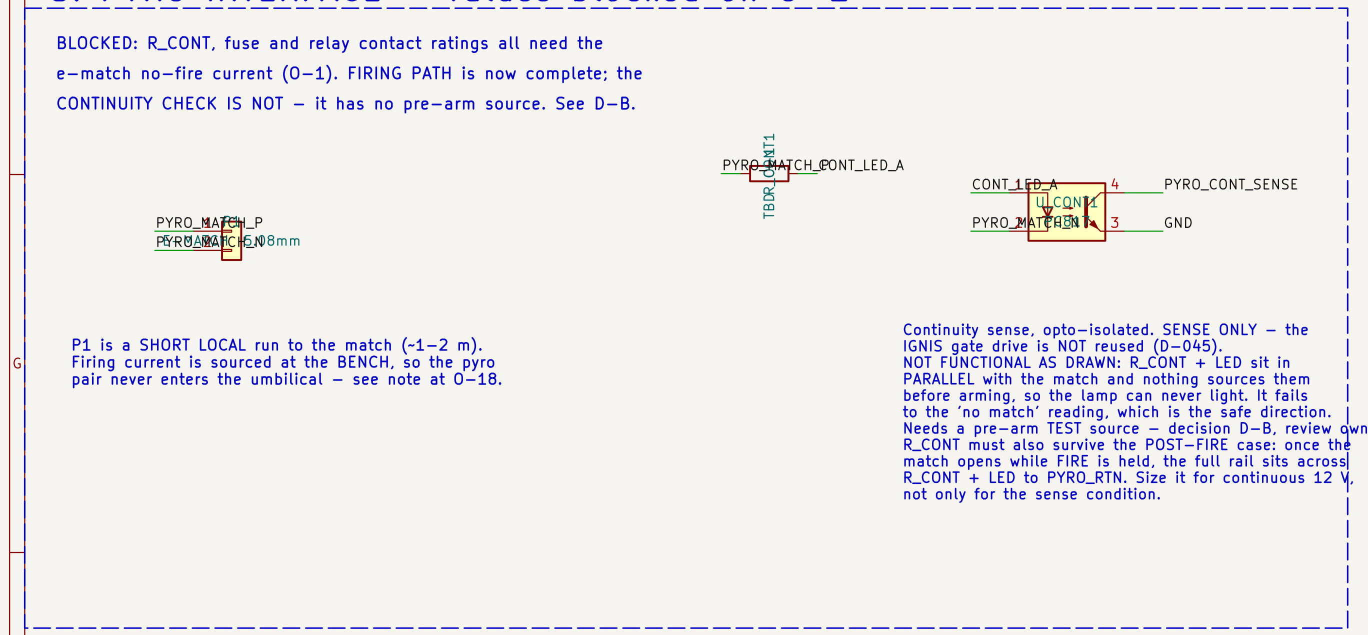
K2 FIRE: energised only while
the FIRE button is held.

K3 SHUNT + LOW SIDE: NC shorts the match
whenever DISARMED (D-008). NO ties the low
side to PYRO_RTN when armed. Break-before-
make REQUIRED – verify per O-3.

J_ARM: withdrawn by hand at the bench,
inserted LAST. Third series element (D-013).

3. PYRO INTERFACE – values blocked on O-1

BLOCKED: R_CONT, fuse and relay contact ratings all need the
e-match no-fire current (O-1). FIRING PATH is now complete; the
CONTINUITY CHECK IS NOT – it has no pre-arm source. See D-B.

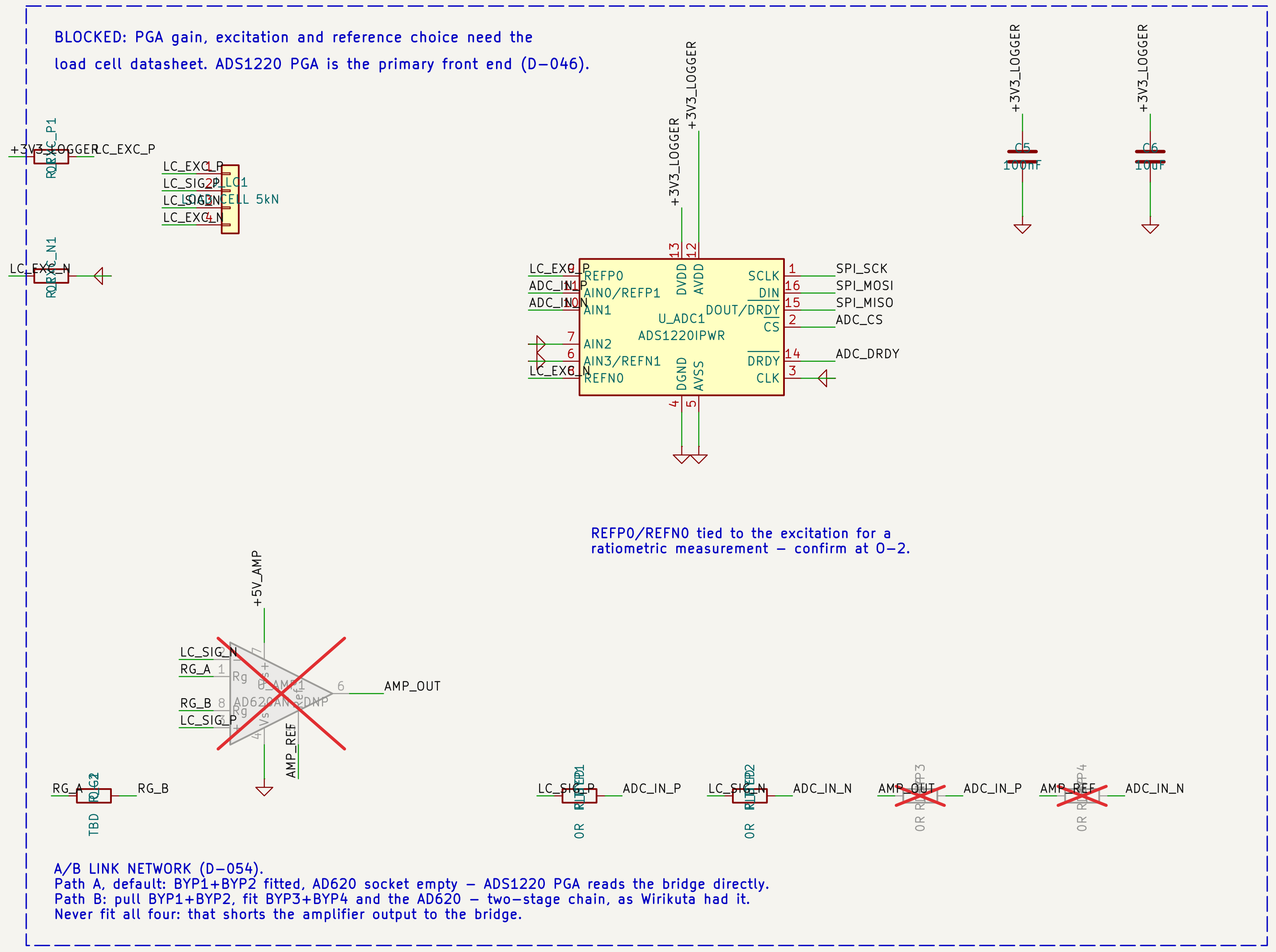

PYRO_MATCH_P
PYRO_MATCH_N
5.08mm
PYRO_MATCH_P
CONT_LED_A
TBD
CONT_LED_A
PYRO_MATCH_N
U_CONT1
PYRO_CONT_SENSE
GND

P1 is a SHORT LOCAL run to the match (~1-2 m).
Firing current is sourced at the BENCH, so the pyro
pair never enters the umbilical – see note at O-18.

Continuity sense, opto-isolated. SENSE ONLY – the
IGNIS gate drive is NOT reused (D-045).
NOT FUNCTIONAL AS DRAWN: R_CONT + LED sit in
PARALLEL with the match and nothing sources them
before arming, so the lamp can never light. It fails
to the 'no match' reading, which is the safe direction.
Needs a pre-arm TEST source – decision D-B, review owner.
R_CONT must also survive the POST-FIRE case: once the
match opens while FIRE is held, the full rail sits across
R_CONT + LED to PYRO_RTN. Size it for continuous 12 V,
not only for the sense condition.

4. ANALOG FRONT END – values blocked on O-2

BLOCKED: PGA gain, excitation and reference choice need the
load cell datasheet. ADS1220 PGA is the primary front end (D-046).


+3V3_LOGGER
LC_EXC_P
LC_EXC_P
LC_SIG_P
LC_SIG_N
LC_EXC_N
LC1
LOAD CELL 5kN
LC_EXC_N
+3V3_LOGGER
+3V3_LOGGER
LC_EXC_P
ADC_IN_P
ADC_IN_N
LC_EXC_N
REFP0
AIN0/REFP1
AIN1
AIN2
AIN3/REFN1
REFN0
DVDD
AVDD
DGND
AVSS
SCLK
DIN
DOUT/DRDY
CS
DRDY
CLK
U_ADC1
ADS1220IPWR
SPI_SCK
SPI_MOSI
SPI_MISO
ADC_CS
ADC_DRDY
+3V3_LOGGER
C5
100nF
+3V3_LOGGER
C6
10uF
+5V_AMP
LC_SIG_N
RG_A
RG_B
LC_SIG_P
U_AMP1
AD620AN
DNP
AMP_OUT
AMP_REF
RG_A
RG_B
TBD
LC_SIG_P
ADC_IN_P
OR
LC_SIG_N
ADC_IN_N
OR
AMP_OUT
ADC_IN_P
OR
AMP_REF
ADC_IN_N
OR

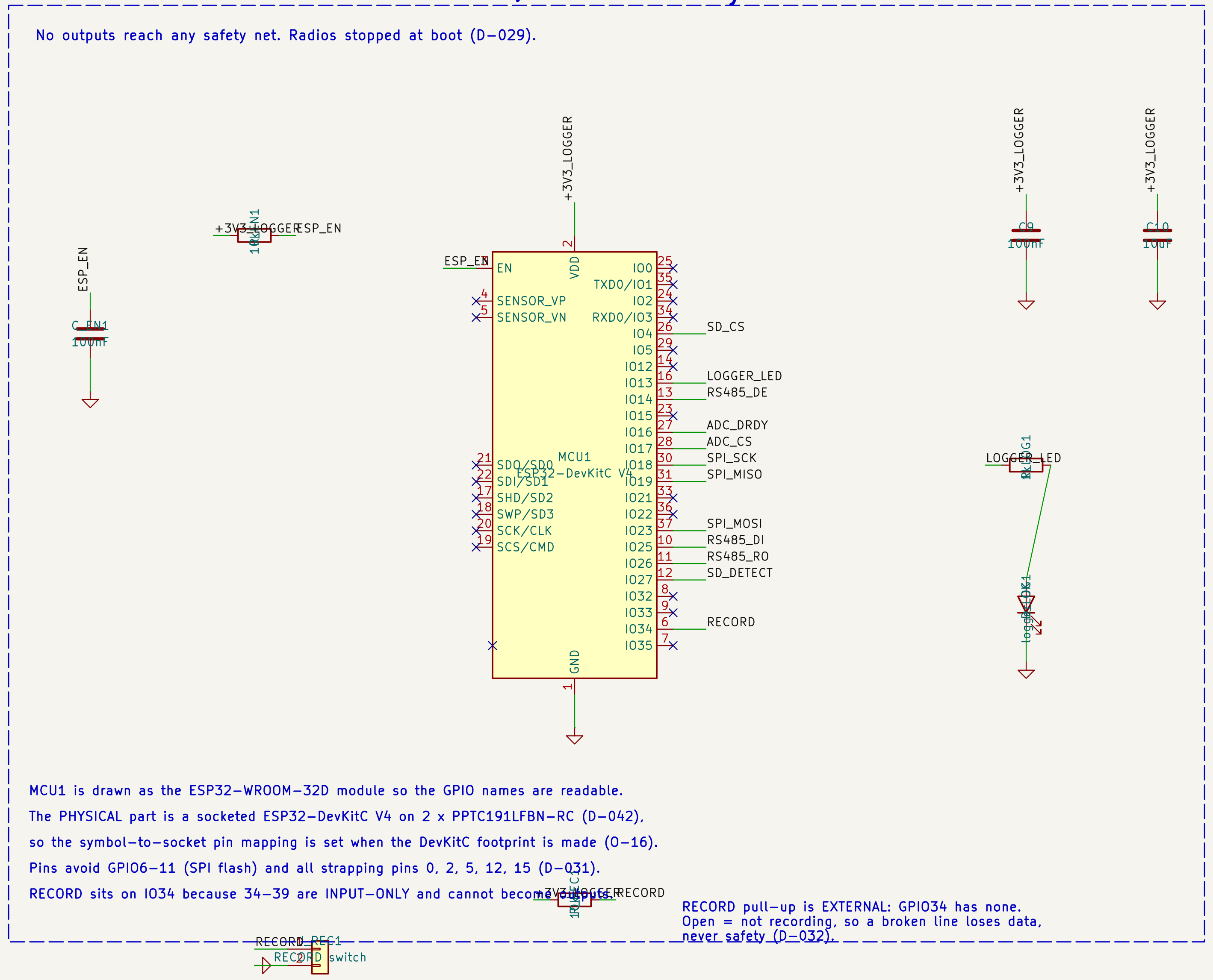
REFP0/REFN0 tied to the excitation for a
ratiometric measurement – confirm at O-2.

A/B LINK NETWORK (D-054).
Path A, default: BYP1+BYP2 fitted, AD620 socket empty – ADS1220 PGA reads the bridge directly.
Path B: pull BYP1+BYP2, fit BYP3+BYP4 and the AD620 – two-stage chain, as Wirikuta had it.
Never fit all four: that shorts the amplifier output to the bridge.

AD620 minimum total supply is 4.6 V, so it CANNOT run on 3.3 V.
Path B therefore needs the +5V_AMP rail, and its output must be kept
inside the ADS1220 input range (O-7). Single-supply use also needs
AMP_REF biased above ground – value TBD with O-2.

5. LOGGER – ESP32-DevKitC V4, zero authority

No outputs reach any safety net. Radios stopped at boot (D-029).


ESP_EN
C_EN1
100nF
+3V3_LOGGER
ESP_EN
10k
+3V3_LOGGER
ESP_EN
EN
SENSOR_VP
SENSOR_VN
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
VDD
GND
MCU1
ESP32-DevKitC V4
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
SD_CS
LOGGER_LED
RS485_DE
ADC_DRDY
ADC_CS
SPI_SCK
SPI_MISO
SPI_MOSI
RS485_DI
RS485_RO
SD_DETECT
RECORD
+3V3_LOGGER
C9
100nF
+3V3_LOGGER
C10
10uF
LOGGER_LED
+3V3_LOGGER
RECORD
RECORD
REC1
RECORD switch

MCU1 is drawn as the ESP32-WROOM-32D module so the GPIO names are readable.

The PHYSICAL part is a socketed ESP32-DevKitC V4 on 2 x PPTC191LFBN-RC (D-042),

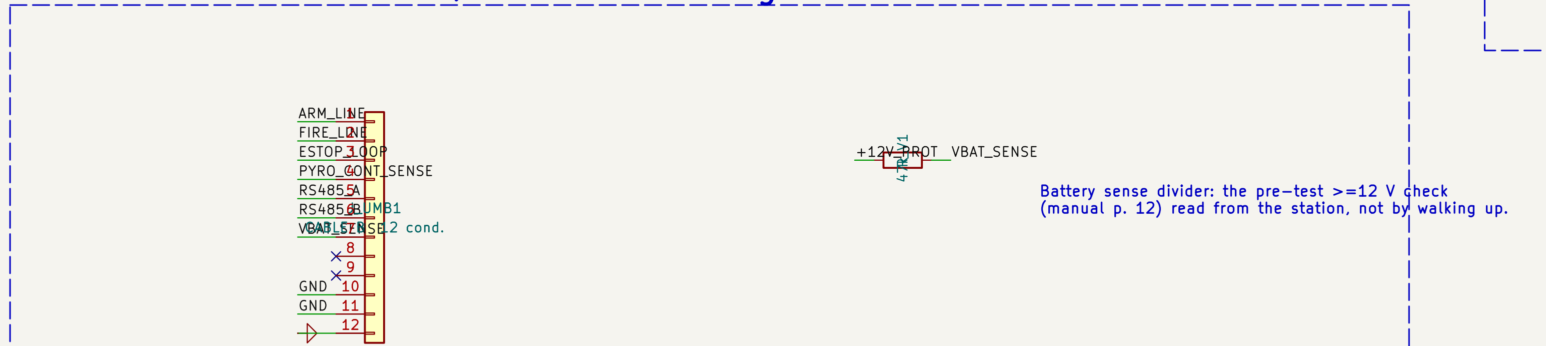
so the symbol-to-socket pin mapping is set when the DevKitC footprint is made (O-16).

Pins avoid GPIO6-11 (SPI flash) and all strapping pins 0, 2, 5, 12, 15 (D-031).

RECORD sits on IO34 because 34-39 are INPUT-ONLY and cannot become outputs.

RECORD pull-up is EXTERNAL: GPIO34 has none.
Open = not recording, so a broken line loses data,
never safety (D-032).

9. UMBILICAL – cable B, control + analog


ARM_LINE
FIRE_LINE
ESTOP_LOOP
PYRO_CONT_SENSE
RS485_A
RS485_B
VBAT_SENSE
J_UMB1
12 cond.
GND
GND
+12V_PROT
VBAT_SENSE
47k
VBAT_SENSE
10k

Battery sense divider: the pre-test >=12 V check
(manual p. 12) read from the station, not by walking up.

PATH 2 DEFERRED FOR REV A – open item O-19. Pins 8 and 9 are RESERVED and

carry explicit no-connect flags. They previously held THRUST_DIFF_P/N labels

that reached nothing on this board: a pair present at both connectors with no

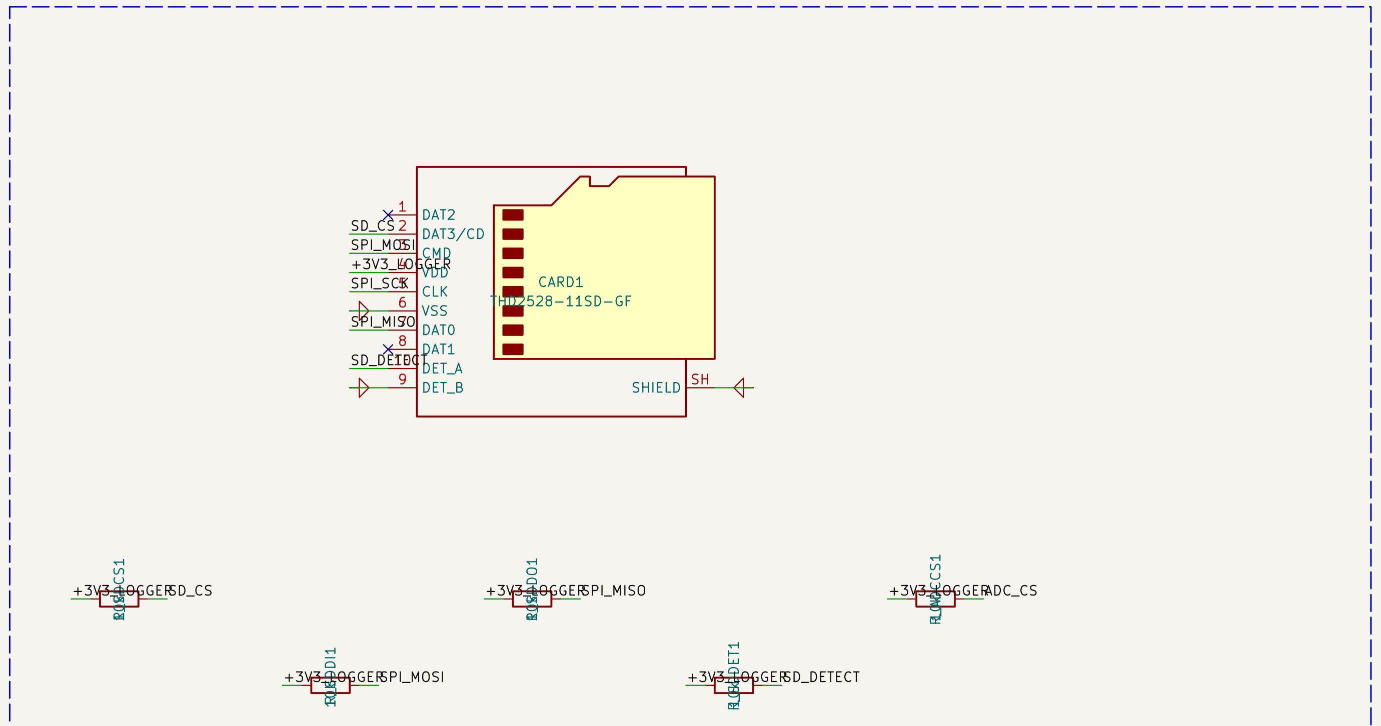
source, which reads as a working analog backup and is not one.

Path 2 needs its OWN buffer – the ADS1220 swallows the signal into SPI.

Loss of path-2 redundancy for Rev A is recorded in REQUIREMENTS.md.

Shield: pin 12 bonds to bench ground HERE ONLY; the station end floats.

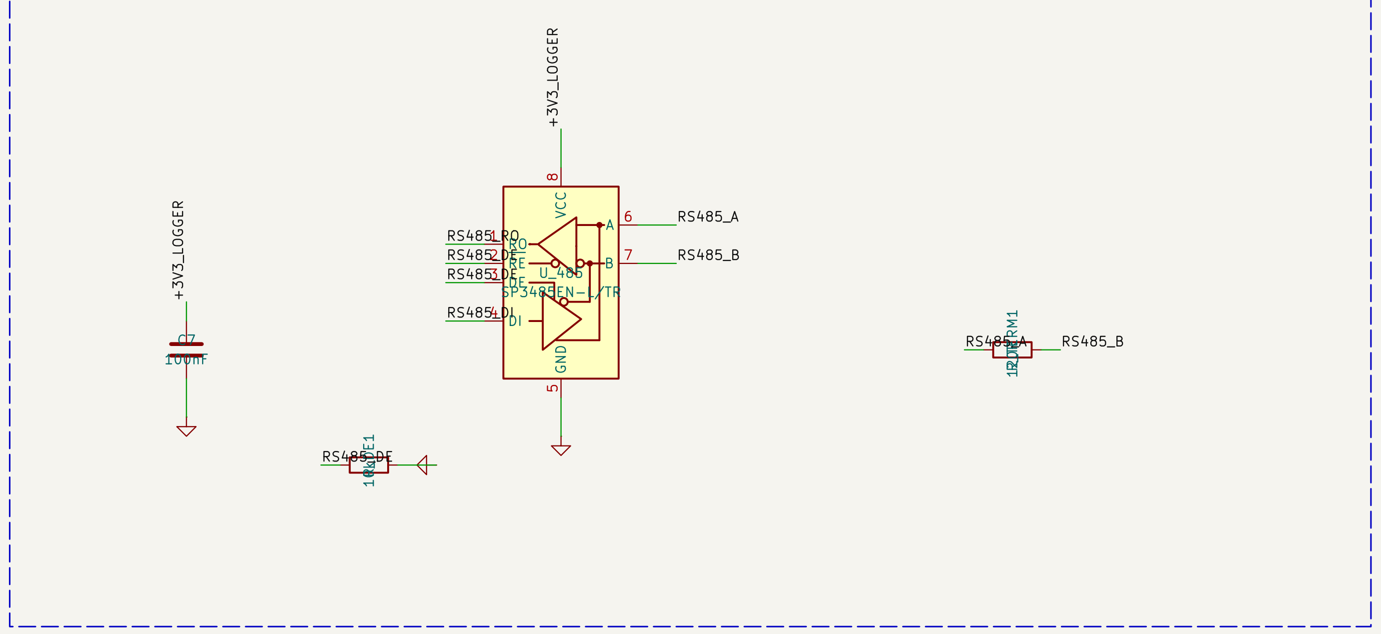
6. STORAGE – on-board microSD


SD_CS
SPI_MOSI
+3V3_LOGGER
SPI_SCK
SPI_MISO
SD_DETECT
DAT2
DAT3/CD
CMD
VDD
CLK
VSS
DAT0
DAT1
DET_A
DET_B
CARD1
THD2528-11SD-GF
SHIELD
SH
+3V3_LOGGER
SD_CS
+3V3_LOGGER
SPI_MISO
+3V3_LOGGER
ADC_CS
+3V3_LOGGER
SPI_MOSI
+3V3_LOGGER
SD_DETECT

Pull-ups per IGNIS v5 Storage.SchDoc.
SD_DETECT on CARD1 pin 10 (DET_A) with a 10k
pull-up; pin 9 (DET_B) to GND. Shell to GND,
pegs drilled 1.3 mm. R_ADCCS added for ADC_CS.

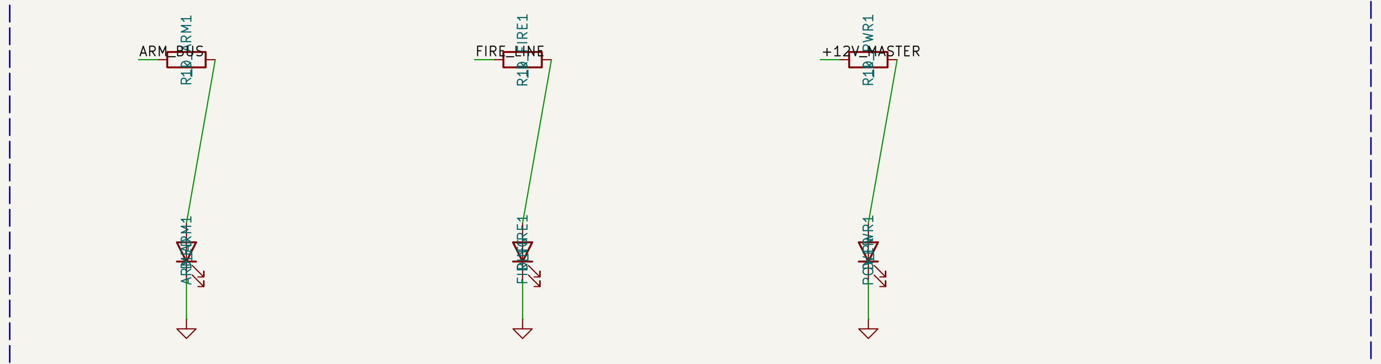
7. TELEMETRY – isolated RS-485 to the station

Data OUTWARD only. Never a command path.


+3V3_LOGGER
C7
100nF
+3V3_LOGGER
RS485_RO
RS485_DE
RS485_DE
RS485_DI
RO
RE
DE
DI
VCC
GND
A
B
U_485
SP3485EN-L/TR
RS485_A
RS485_B
RS485_DE
10k
RS485_A
RS485_B
R_TERM1

8. INDICATORS – beacon driven from the REAL arm/fire lines

D-017: the beacon must not be able to misreport state,
so it is NOT driven by the MCU. WS2812B cannot serve here (D-043).


ARM_BUS
R_ARM1
FIRE_LINE
R_FIRE1
+12V_MASTER
R_PWR1

Beacon LED part is TBD: must be visible at 20-40 m (manual p. 24).

TBD values await O-1 (e-match no-fire current) and O-2 (load cell datasheet).

Review owner: Emilio Guadarrama. Generated by hardware/tools/gen_bench.py
Igniter safe state is SHORTED via K3 normally-closed. Values marked TBD await O-1 / O-2
PRELIMINARY – NOT APPROVED. No MCU on the arm, fire or e-stop path.
Ignitia Rocket Lab
Sheet: /
File: talavera-bench.kicad_sch
Title: TALAVERA v1 – Bench Board
Size: A2
Date: 2026-11-06
Rev: A
KiCad E.D.A. 10.0.5
Id: 1/1